

MICRO GESTURE RECOGNITION WITH TERAHERTZ RADAR BASED ON DIAGONAL PROFILE OF RANGE-DOPPLER MAP

Xing Wang, Rui Min, Zongyong Cui, Zongjie Cao

School of Information and Communication Engineering
University of Electronic Science and Technology of China
Chengdu, China 611731

ABSTRACT

Gestures can be gradually used to achieve natural and direct communication between people and machines, not limited to people. However, micro gesture motion sensing using traditional sensors and techniques is challenging because of the difference and diversity of finger motion. This paper proposes a novel method for micro gesture recognition based on the high resolution of terahertz radar, which can capture fine changes during gesture movement. From these radar echoes collected in gesture motion, the low dimensional projection features of diagonal profile based on range-Doppler map (RDM) are extracted to characterize the motion difference between gestures. Then, a convolutional neural network (CNN) is employed to realize the gesture recognition system. The experimental results show that the proposed method can achieve a high recognition rate of 98.06% with six dynamic micro gestures across eight participants.

Index Terms— micro gesture recognition, terahertz radar, high resolution, diagonal profile

1. INTRODUCTION

Compared to a graphical user interface, natural user interfaces implement human-computer interaction in a direct and near-realistic way [1]. One of the important categories is gesture-based interaction. It regards the user as an input device and realizes the natural and efficient human-computer interaction based on the daily habits and behavior cognition of the user.

Radar is originally used for target detection and ranging, and gradually expanded to military, transportation, meteorology and other fields. For gesture recognition, radar has attracted attention due to its unique advantages, such as robustness to environmental changes, avoidance of privacy leakage, and flexibility in settings. Therefore, it can be commonly used in smart home, game interaction, industrial control, intelligent driving, and other applications.

Currently, a 25Ghz radar prototype capable of capturing three-dimensional coordinates and speed of gestures was first designed, which proved the feasibility of using radar for gesture recognition [2]. Then, Project Soli developed by Google

realized the end-to-end millimeter-wave radar gesture recognition system from the bottom hardware to the top application [3]. Additionally, Texas Instruments also correspondingly implemented multi-gesture real-time classification on its AWR1642 radar chip, and overcomes the limitations of memory, processing capabilities and angular resolution that exist in traditional radar applications [4].

Based on the above hardware foundation or other radar systems, various research methods have also been proposed for gesture recognition. Recently, CNN and recurrent neural network (RNN) are combined to directly extract the RDM sequence features generated by the motion of gesture, which can classify 11 dynamic gestures with the recognition rates of 87% [5]. Besides, the sparse time-frequency features contained in dynamic gestures are extracted and sent to the SVM classifier for gesture recognition [6]. Furthermore, the connectionist temporal classification (CTC) algorithm is applied to predict different gesture labels from unsegmented input streams, so as to realize low delay of gesture recognition [7].

At present, gesture recognition based on radar has a further development in hardware and algorithm, but it is always a challenge to accurately recognize the micro gesture. These gestures are mainly composed of motion changes of multiple fingers and corresponding joints, with small movement and short duration. Limited by signal bandwidth and carrier frequency, traditional radars cannot perceive small changes of multiple scattering centers on hand. However, the terahertz radar with its high resolution in time and space has the ability to capture small movements. On this basis, there are considerable differences in micro gestures performed by different people. Traditional methods cannot effectively extract the motion features of micro gestures to achieve the accuracy of classification. At the same time, it may be accompanied by high data loads and calculations, which is not suitable for some practical mobile applications.

Aiming at the problems raised above, this paper relies on a high-resolution terahertz radar to capture the motion changes of multiple scattering centers on hand. Based on the low-dimensional projection of RDM, the features of diagonal profile are then extracted to recognize different micro gestures.

2. TERAHERTZ RADAR SYSTEM

The structure of the terahertz radar system is illustrated in Fig. 1. It operates at a central frequency of 220 GHz. The transmitted signal is based on a FMCW modulation scheme that its frequency changing with time. This sweep in frequency is commonly referred to as a chirp. For the terahertz radar system, the chirp bandwidth is 12.8 GHz and chirp time is 1ms. Therefore, this radar system has a high resolution in range and velocity, which can be respectively represented as follows:

$$r_{res} = \frac{c}{2B} \approx 1 \text{ cm}, \quad (1)$$

$$v_{res} = \frac{c}{2f_c N T_c} \approx 2.13 \text{ cm/s}, \quad (2)$$

where c is the speed of light, and B is the chirp bandwidth. f_c and T_c are the central frequency and the chirp time separately. N is the number of accumulated echoes which is set to 32.

Therefore, as a non-rigid object, multiple scattering centers of hand in the process of moving can be sensed by the terahertz radar from two aspects of range and velocity. This makes the difference of movement between different gestures more obvious, and micro gesture recognition is practicable.

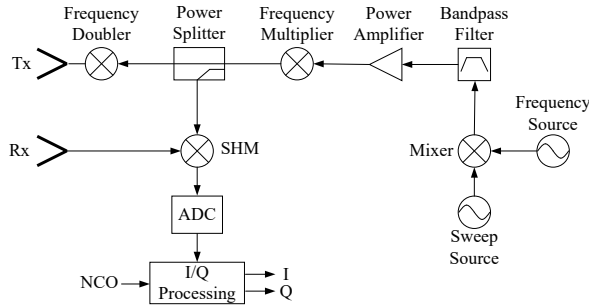


Fig. 1. The structure of the terahertz radar system.

3. METHOD OF GESTURE RECOGNITION

The received signal $s(t, T)$ from the terahertz radar includes two distinct time dimensions, where t is the sampling on a single echo along fast time, and T is the index of echoes collected over slow time. This 2D signal provides information about the instantaneous range and velocity change of multiple scattering centers on the hand, which can be then used to recognize micro gestures.

3.1. Data preprocessing

For data streams received from the terahertz radar, static interference and background noise are inevitably introduced by other objects. Especially for micro gestures, the energy of echo signal is relatively weak, and it is easy to submerge in the complex environment. At the same time, the start and end

time of hand motion also needs to be detected, so as to extract features for recognition later. Therefore, it is necessary to preprocess the radar echoes effectively.

In this paper, a set of radar echoes form a data frame and this can be used as the observation window for radar signal processing. The 1D FFT along fast time is applied to obtain high range resolution profile (HRRP) sequence frame by frame. On this basis, the static interference can be suppressed by removing the low-frequency components after using FFT along the slow time. Then, the statistical results of amplitude with background noise approximately follow Rayleigh distribution [8], i.e.

$$p(x) = \frac{x}{\mu^2} \exp\left(-\frac{x^2}{2\mu^2}\right), \quad (3)$$

where $x \geq 0$, and parameter μ can be estimated based on the mean of x . From it, background noise can be reduced by CFAR detector with threshold as follows:

$$Th = \mu \sqrt{2 \ln\left(\frac{1}{P_{fa}}\right)}, \quad (4)$$

where P_{fa} is false alarm probability. After the above processing, the residual isolated noise points can be removed by singular value decomposition [9].

Next, the start and end time of gestures in a continuous data stream can be detected with the energy change based on RDM. The RDM is obtained by applying a 1D FFT along slow time with HRRP sequence in a frame. The instantaneous energy value calculated by summing the amplitude of RDM for each frame is spliced into a fluctuating curve over time [10]. Using a moving average for the energy curve, its points of minimal value correspond to the start and end time of adjacent gestures. Fig. 2 shows the result of data preprocessing.

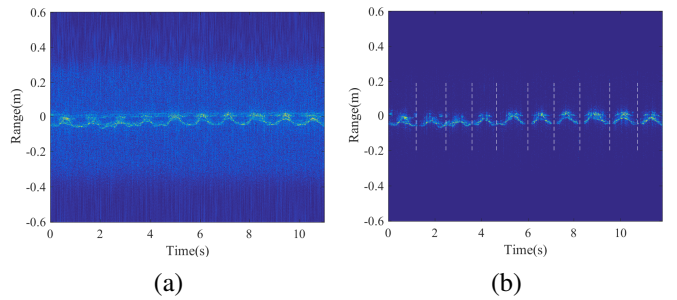


Fig. 2. Data preprocessing is performed on raw data streams. (a) HRRP sequence before preprocessing. (b) HRRP sequence after preprocessing with gesture segmentation.

3.2. Feature extraction based on diagonal profile

The detected target is recorded as a peak in the RDM, and the position of a peak corresponding to the fast and slow time

axes reflects the range and velocity of the target relative to radar, respectively. Due to the high resolution of the terahertz radar, RDM sequence can capture subtle changes in range, velocity, and trajectory of multiple scattering centers on hand. However, there is data redundancy using RDM sequence as input for gesture recognition directly, resulting in the overhead of calculation and storage. This is not suitable for some practical mobile application scenarios. Therefore, this paper extracts the features of diagonal profile based on RDM to characterize the motion of micro gesture. Fig. 3 shows the process.

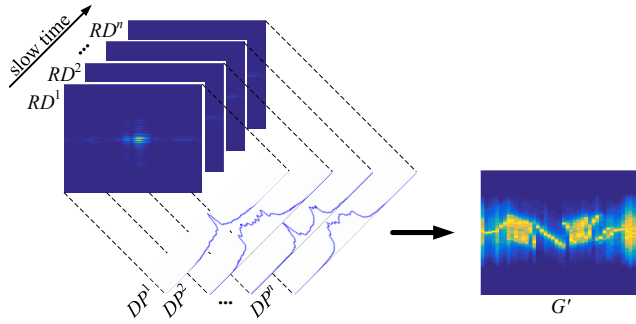


Fig. 3. The motion of micro gesture is characterized by diagonal profile sequence.

After data preprocessing, the normalized RDM sequence corresponding to a complete gesture can be represented as follows:

$$G = (RD^1, RD^2, \dots, RD^n), \quad (5)$$

where n is the number of frames include in a gesture from start to end. RD^i is the i -th RDM with size $K \times L$, i.e.

$$RD^i = \begin{pmatrix} rd_{11}^i & \dots & rd_{1L}^i \\ \vdots & \ddots & \vdots \\ rd_{K1}^i & \dots & rd_{KL}^i \end{pmatrix}. \quad (6)$$

The diagonal profile, DP^i , based on the i -th RDM is defined as follows:

$$DP^i = (dp_{-K+1}^i, dp_{-K+2}^i, \dots, dp_{L-1}^i)^T, \quad (7)$$

$$dp_j^i = \sum_{l-k=j} rd_{kl}^i, \quad (8)$$

where $j = -K + 1, -K + 2, \dots, L - 1$. The diagonal profile is a low-dimensional projection of RDM along its diagonal, which shows the energy distribution reflected by multiple scattering centers in a low amount of data. Furthermore, the diagonal profile avoids the overlap of projection for multiple scattering centers in the two main feature dimensions of range and velocity, improving the resolution of gesture description.

Finally, the full features of gesture motion are obtained by stitching the diagonal profile sequence along slow time, i.e.

$$G' = (DP^1, DP^2, \dots, DP^n). \quad (9)$$

3.3. Gesture classification using CNN

In this paper, the extracted features are used as input to CNN for gesture classification. Considering the difference in gesture duration, the feature G' is processed by nearest neighbor interpolation to obtain the same input size. Then, the input data is enhanced by column normalization of features to improve the generalization ability of the model. Finally, the classification model is trained and outputs gesture labels with the architecture of CNN shown in Fig. 4.

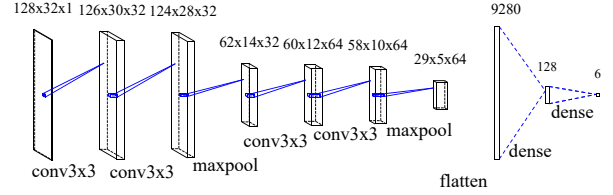


Fig. 4. The architecture of CNN.

4. EXPERIMENT AND EVALUATION

To verify the effectiveness of the proposed method, this paper defines six micro gestures shown in Fig. 5 for recognition, including (a) beckon, (b) circle, (c) grasp, (d) left and right, (e) forward and backward, (f) click. Besides, the data set constructed in this paper was completed by 8 users, each of whom repeated 30 times for each type of gesture, resulting in $6 \times 8 \times 30 = 1440$ samples.

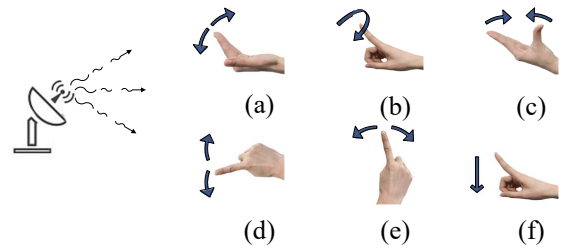


Fig. 5. Six gestures to be recognized.

Next, the relevant settings in this experiment are described in detail. First, the number of echoes contained in a frame and false alarm probability during preprocessing are set to 32 and 0.001, respectively. Second, the output size of RDM is set to 512×128 after windowing and FFT. Last, the extracted feature is fed to CNN with a size of 128×32 . To obtain as much effective information as possible from limited samples and reduce the occurrence of over-fitting to a certain extent, 7-fold cross-validation is used to evaluate the overall performance of gesture recognition model.

On this basis, we conduct comparison experiments on Doppler profile and the proposed method to evaluate the ef-

Table 1. Recognition rate per each fold in cross-validation.

Methods	Fold1	Fold2	Fold3	Fold4	Fold5	Fold6	Fold7	Avg
Doppler profile	91.75%	92.23%	91.15%	94.27%	93.72%	95.29%	93.68%	93.16%
The proposed method	98.45%	97.93%	96.88%	98.96%	97.38%	98.95%	97.89%	98.06%

fect of diagonal profile. Doppler profile is the projection of RDM along velocity axis. Table 1 shows the results of 7-fold cross-validation. The proposed method in this paper achieves a high average recognition rate of 98.06%, which is higher than the results based on Doppler profile.

Furthermore, the confusion matrix of CNN classifier shown in Fig. 6. It can be seen that the model can recognize each class of gestures well.

Confusion Matrix

True label	Beckon	0.97	0.00	0.01	0.00	0.01	0.00
	Circle	0.00	0.97	0.00	0.03	0.00	0.00
	Grasp	0.00	0.00	1.00	0.00	0.00	0.00
	LR	0.00	0.00	0.00	1.00	0.00	0.00
	FB	0.00	0.00	0.00	0.00	1.00	0.00
	Click	0.00	0.00	0.00	0.06	0.00	0.94
		Predicted label					
		Beckon	Circle	Grasp	LR	FB	Click

Fig. 6. The confusion matrix of CNN classifier.

5. CONCLUSION

In this paper, a novel micro gesture recognition method based on high-resolution terahertz radar is proposed, and its recognition performance for six gestures across eight participants is evaluated. After radar echoes are preprocessed, the diagonal profile sequence of RDM is extracted and then sent to the CNN classifier for gesture classification. The high recognition rate of 98.06% obtained in the experiment proves that the features of diagonal profile can effectively characterize gesture motion with a low amount of data.

6. REFERENCES

[1] A. Subramanian, "Integration of natural user interface in a real-world environment," in *2015 IEEE International*

Conference on Computational Intelligence Communication Technology, Feb 2015, pp. 714–718.

[2] P. Molchanov, S. Gupta, K. Kim, and K. Pulli, "Short-range fmcw monopulse radar for hand-gesture sensing," in *2015 IEEE Radar Conference (RadarCon)*, May 2015, pp. 1491–1496.

[3] Jaime Lien, Nicholas Gillian, M. Karagozler, Patrick Amihoud, Carsten Schwesig, Erik Olson, Hakim Raja, and Ivan Poupyrev, "Soli: Ubiquitous gesture sensing with millimeter wave radar," *ACM Transactions on Graphics*, vol. 35, pp. 1–19, 07 2016.

[4] P. Goswami, S. Rao, S. Bharadwaj, and A. Nguyen, "Real-time multi-gesture recognition using 77 ghz fmcw mimo single chip radar," in *2019 IEEE International Conference on Consumer Electronics (ICCE)*, Jan 2019, pp. 1–4.

[5] Saiwen Wang, Jie Song, Jaime Lien, Ivan Poupyrev, and Otmar Hilliges, "Interacting with soli: Exploring fine-grained dynamic gesture recognition in the radio-frequency spectrum," 10 2016, pp. 851–860.

[6] G. Li, R. Zhang, M. Ritchie, and H. Griffiths, "Sparsity-based dynamic hand gesture recognition using micro-doppler signatures," in *2017 IEEE Radar Conference (RadarConf)*, May 2017, pp. 0928–0931.

[7] Z. Zhang, Z. Tian, and M. Zhou, "Latern: Dynamic continuous hand gesture recognition using fmcw radar sensor," *IEEE Sensors Journal*, vol. 18, no. 8, pp. 3278–3289, April 2018.

[8] Victor C Chen and Shie Qian, "Cfar detection and extraction of unknown signal in noise with time-frequency gabor transform," in *Wavelet Applications III*. International Society for Optics and Photonics, 1996, vol. 2762, pp. 285–294.

[9] S. K. Jha and R. D. S. Yadava, "Denoising by singular value decomposition and its application to electronic nose data processing," *IEEE Sensors Journal*, vol. 11, no. 1, pp. 35–44, Jan 2011.

[10] J. Choi, S. Ryu, and J. Kim, "Short-range radar based real-time hand gesture recognition using lstm encoder," *IEEE Access*, vol. 7, pp. 33610–33618, 2019.